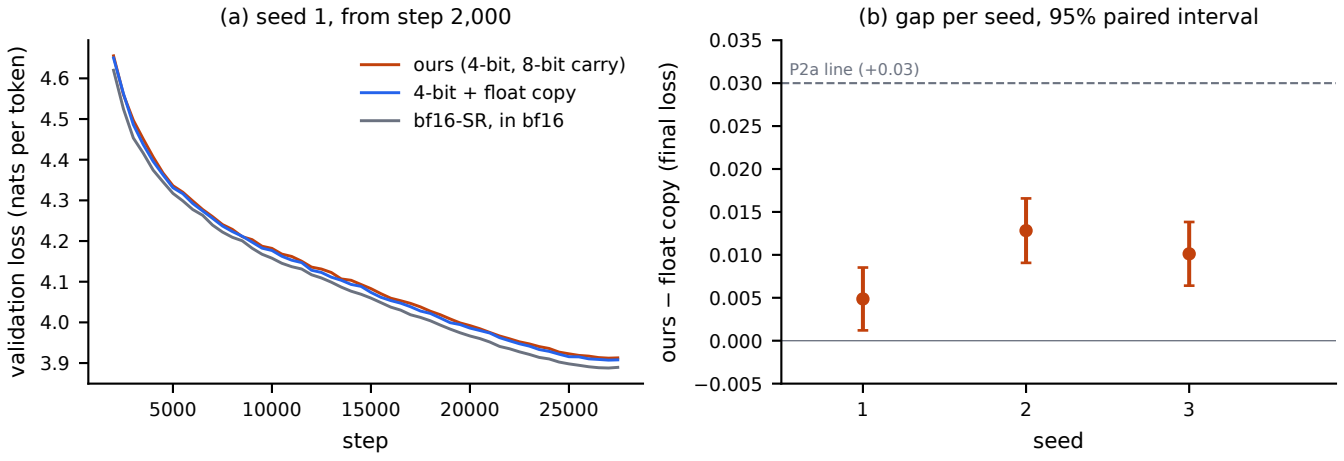
(a) seed 1, from step 2,000
validation loss (nats per token)
step
ours (4-bit, 8-bit carry)
4-bit + float copy
bf16-SR, in bf16
4.6
4.5
4.4
4.3
4.2
4.1
4.0
3.9
5000
10000
15000
20000
25000
(b) gap per seed, 95% paired interval
ours − float copy (final loss)
seed
P2a line (+0.03)
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
−0.005
1
2
3